

TRANSFORMER ARCHITECTURE: MULTI-HEAD SELF-ATTENTION MECHANISM



LAYMAN TERM

Imagine a big team of specialized spotters looking at a complex scene but focusing on different details simultaneously, then sharing their views to create a complete understanding. Multi-head self-attention works similarly in Transformers by letting multiple "attention heads" analyze different relationships in data at once, giving engineers powerful insight into complex input.



DEFINITION

Multi-head self-attention is a parallelized attention mechanism in Transformer models that computes multiple attention distributions simultaneously to capture diverse patterns from the same input sequence.



WHEN TO USE

Multi-head self-attention is essential in processing sequence data where relationships among elements vary contextually, such as in natural language processing, for tasks like translation, summarization, and generating contextual embeddings.



HOW TO MAKE

- Tokenize input sequence into embeddings.
- Create Query (Q), Key (K), and Value (V) matrices by linear projection of embeddings for each head.
- Compute scaled dot-product attention independently for each head: $\text{softmax}(QK^T / \sqrt{d_k})V$.
- Concatenate outputs from all attention heads.



TYPES

- Scaled Dot-Product Attention
- Single-Head Self-Attention
- Multi-Head Self-Attention
- Masked Multi-Head Attention (used in decoder for autoregression)
- Cross-Attention (encoder-decoder attention)



POINTS TO PONDER

- Computational cost increases linearly with the number of attention heads and quadratic with sequence length.
- Attention heads may learn redundant patterns if not regularized properly.
- Positional encoding is necessary as self-attention itself is permutation invariant.
- Multi-head attention may struggle with very long sequences without adaptations like sparse or efficient attention variants.
- Balancing number of heads and embedding dimensions impacts model capacity and speed.

- Pass concatenated output through a final linear projection.
- Integrate residual connections and layer normalization.
- Feed the result to subsequent Transformer layers for further processing.

CONCLUSION

Multi-head self-attention enables Transformers to capture multiple distinct relationships within the same sequence simultaneously. Its core formula is $\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$, where $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$.

RECOMMENDED SOURCES

- <https://www.geeksforgeeks.org/nlp/multi-head-attention-mechanism/>
- <https://www.datacamp.com/tutorial/multi-head-attention-transformers>
- https://d2l.ai/chapter_attention-mechanisms-and-transformers/transformer.html

BEST VIDEO GUIDE

RECOMMENDED CONTENT:

Multi-Head Self-Attention Transformer Microsoft Mechanics

LINK: [HTTPS://YOUTUBE.COM/WATCH?V=IDULHQ2PRO](https://youtube.com/watch?v=IDULHQ2PRO)